Preprint

SUPPLEMENTARY MATERIAL

The Two-Potential Energy of the Constitutive Law $\mu(x) = x/(x+1)$ in Bimetric MOND

Stephen Nelson

Field Dynamics Research
stephen@fielddynamics.org

Abstract
Milgrom's bimetric MOND (BIMOND) has two free choices in its nonrelativistic limit: the ratio of the two Einstein–Hilbert couplings α, β and an interaction function $\mathcal{M}$, which is left unspecified apart from its two limits. We show that the difference potential $\phi^* = \phi - \hat{\phi}$ obeys a Poisson equation independent of $\mathcal{M}$ if and only if $\alpha + \beta = 0$, and that in the tetrad form of the Einstein–Hilbert term a frame on the second root of the tetrad equation enters with the opposite sign, so equal couplings give exactly this case. Requiring that BIMOND realise the constitutive law $\mu(x) = x/(x+1)$ then fixes $\mathcal{M}$ uniquely: $\mathcal{M} = 2x - 1 + 2b - b^2 + 2\ln b$, with $b = 1/(1+x)$ and x the total field in units of a_0. It meets Milgrom's two conditions exactly. It is the Legendre dual of the AQUAL kinetic function F, with the logarithms of the two cancelling in the dual pairing, and its energy is strictly convex, with a Hessian ratio that is the reciprocal of the AQUAL one.

Keywords: BIMOND, QUMOND, AQUAL, MOND, bimetric gravity, Legendre duality, proof

Setting

The nonrelativistic limit of BIMOND [1, 2] has two potentials: ϕ, felt by matter of density ρ, and $\hat{\phi}$, felt by twin matter of density $\hat{\rho}$. With $\phi^* = \phi - \hat{\phi}$ and $z = |\nabla\phi^*|^2/a_0^2$ the Lagrangian density is

$$\mathcal{L} = -\frac{1}{8\pi G}\Big[\beta|\nabla\phi|^2 + \alpha|\nabla\hat{\phi}|^2 - a_0^2\,\mathcal{M}(z)\Big] - \rho\,\phi + \hat{\rho}\,\hat{\phi}, \tag{1}$$

with the twin sign of Ref. [2]. Its Euler–Lagrange equations are

$$\begin{aligned}\nabla\cdot\big[\beta\nabla\phi - \mathcal{M}'\nabla\phi^*\big] &= 4\pi G\rho,\\ \nabla\cdot\big[\alpha\nabla\hat{\phi} + \mathcal{M}'\nabla\phi^*\big] &= -4\pi G\hat{\rho},\end{aligned} \tag{2}$$

where $\mathcal{M}' = d\mathcal{M}/dz$. An additive constant in $\mathcal{M}$ leaves Eq. (2) unchanged, and we fix it by $\mathcal{M}(0) = 0$. Throughout,

$$\mu(x) = \frac{x}{x+1}. \tag{3}$$

The Coupling Ratio

Theorem 1. *Let $\alpha\beta \neq 0$ and let $\mathcal{M}'$ be nonconstant. The potential ϕ^* obeys a linear equation independent of $\mathcal{M}$ if and only if $\alpha + \beta = 0$. With $\beta = 1$ the system is then*

$$\begin{aligned}\Delta\phi^* &= 4\pi G\,(\rho - \hat{\rho}),\\ \Delta\phi &= 4\pi G\rho + \nabla\cdot\big(\mathcal{M}'\nabla\phi^*\big).\end{aligned} \tag{4}$$

Proof. α times the first of Eqs. (2) minus β times the second gives

$$\nabla\cdot\Big[\big(\alpha\beta - (\alpha+\beta)\mathcal{M}'\big)\nabla\phi^*\Big] = 4\pi G(\alpha\rho + \beta\hat{\rho}), \tag{5}$$

the unique combination, up to a factor, in which $\hat{\phi}$ cancels. Its coefficient is a constant exactly when $(\alpha+\beta)\mathcal{M}'$ is, that is, when $\alpha + \beta = 0$. Setting $\alpha = -1$, $\beta = 1$ in Eq. (5) and in the first of Eqs. (2) gives Eq. (4). □

A MOND theory needs a nonconstant $\mathcal{M}'$, since it must tend to 0 at high acceleration and diverge at low acceleration. For $\alpha + \beta = 0$, Eq. (4) is the quasilinear formulation QUMOND [3]: two Poisson equations, the second sourced by the first.

Lemma 1 (orientation). *Write the Einstein–Hilbert term of a frame ${e^a}_\mu$ in its tetrad form, $S[e] = (16\pi G)^{-1}\int \det(e)\,R[g(e)]\,d^4x$ with $g = e^T\eta e$, the 4-form $\epsilon_{abcd}\,e^a\wedge e^b\wedge R^{cd}$ up to normalisation. For constant $\Lambda \in O(1,3)$, $S[\Lambda e] = \det(\Lambda)\,S[e]$. In particular spatial parity P gives $S[Pe] = -S[e]$.*

Proof. $(\Lambda e)^T\eta(\Lambda e) = e^T\Lambda^T\eta\Lambda e = e^T\eta e$, so g and $R[g]$ are unchanged, while $\det(\Lambda e) = \det(\Lambda)\det(e)$. Equivalently, in the 4-form, $\epsilon_{abcd}\Lambda^a{}_{a'}\Lambda^b{}_{b'}\Lambda^c{}_{c'}\Lambda^d{}_{d'} = \det(\Lambda)\,\epsilon_{a'b'c'd'}$. □

The tetrad equation $g = e^T\eta e$ has two roots, $\det e = \pm\sqrt{-g}$, exchanged by P [6]. If the second metric $\hat{g}$ is carried by a frame on the negative root and both frames have the same coupling $(16\pi G)^{-1}$, then $\det(\hat{e})\,\hat{R} = -\sqrt{-\hat{g}}\,\hat{R}$, so in BIMOND's metric notation $\beta = 1$ and $\alpha = -1$: the case of Theorem 1.

The Interaction Function

Definition. *BIMOND realises μ if, for $\hat{\rho} = 0$, its spherically symmetric fields coincide with those of AQUAL [4] with constitutive law μ.*

Proposition 1. *Write $x = |\nabla\phi|/a_0$ for the total field and $y = |\nabla\phi^*|/a_0$. BIMOND with $\alpha + \beta = 0$ realises μ if and only if $1 + \mathcal{M}'(y^2) = \nu(y)$, where $\nu(y) = x/y$ and $y = x\,\mu(x)$. For Eq. (3),*

$$x(y) = \frac{y + \sqrt{y^2 + 4y}}{2}, \qquad \nu(y) = \frac{1}{2} + \sqrt{\frac{1}{4} + \frac{1}{y}}. \tag{6}$$

Proof. In spherical symmetry Gauss's law applied to Eq. (4) gives $|\nabla\phi^*| = g_N$, the Newtonian field, and $|\nabla\phi| = (1 + \mathcal{M}')\, g_N$. AQUAL gives $\mu(|\nabla\phi|/a_0)\,|\nabla\phi| = g_N$. These agree for every g_N exactly when $1 + \mathcal{M}'$ is the inverse partner ν. Since $y = x^2/(1 + x)$ increases from 0 to ∞, the inverse is unique and is Eq. (6). □

Theorem 2. *The unique $\mathcal{M}$ with $1 + \mathcal{M}'(y^2) = \nu(y)$ and $\mathcal{M}(0) = 0$ is, with $b = 1/(1 + x)$,*

$$\boxed{\mathcal{M} = 2x - 1 + 2b - b^2 + 2\ln b,} \tag{7}$$

equivalently $\mathcal{M} = x(2x^2 + 3x + 2)/(x + 1)^2 - 2\ln(1 + x)$, or, in y,

$$\mathcal{M} = \tfrac{1}{2}\sqrt{y}\,(y + 2)\sqrt{y + 4} - \tfrac{1}{2}y^2 - 4\,\mathrm{arsinh}\big(\tfrac{1}{2}\sqrt{y}\big). \tag{8}$$

Its derivative is $\mathcal{M}'(z) = -\frac{1}{2} + \sqrt{\frac{1}{4} + z^{-1/2}}$.

Proof. Two such functions differ by a constant, which $\mathcal{M}(0) = 0$ removes. Since $x - y = x/(1 + x) = \mu(x)$,

$$\frac{d\mathcal{M}}{dy} = 2y\,\mathcal{M}'(y^2) = 2(x - y) = 2\,\mu(x). \tag{9}$$

With $dy/dx = x(x + 2)/(x + 1)^2$ and $t = s + 1$,

$$\begin{aligned}\mathcal{M} &= 2\int_0^x \frac{s^2(s + 2)}{(s + 1)^3}\,ds \\ &= 2\int_1^{1+x} \Big(1 - \frac{1}{t} - \frac{1}{t^2} + \frac{1}{t^3}\Big)dt,\end{aligned} \tag{10}$$

which integrates to Eq. (7). Eq. (8) has the same derivative $2(x - y)$ in y and vanishes at $y = 0$. □

Corollary 1 (the two limits). *As $z \to \infty$, $\mathcal{M}'(z) \to 0$; as $z \to 0$, $\mathcal{M}'(z) = z^{-1/4} - \frac{1}{2} + \frac{1}{8}z^{1/4} + O(z^{3/4})$. Correspondingly*

$$\begin{aligned}\mathcal{M} &= \tfrac{4}{3}y^{3/2} - \tfrac{1}{2}y^2 + \tfrac{1}{10}y^{5/2} + O(y^3), \\ &\qquad y \to 0, \\ \mathcal{M} &= 2y - 2\ln y + 1 - \frac{4}{y} + \frac{5}{y^2} + O(y^{-3}), \\ &\qquad y \to \infty.\end{aligned} \tag{11}$$

Proof. Expand Eq. (6) and Eq. (8). □

These are exactly the two conditions BIMOND imposes on $\mathcal{M}$ [1]: $\mathcal{M}' \to 0$ in the Newtonian regime and $\mathcal{M}' \approx z^{-1/4}$ in the deep-MOND regime, here with unit coefficient and no free normalisation.

Legendre Duality and Convexity

Let $g(x) = F(x^2) = x^2 - 2x + 2\ln(1 + x)$ be the AQUAL kinetic function of μ [5], and $Q(y) = y^2 + \mathcal{M}$ the kinetic function of Eq. (4): the source of $\Delta\phi$ is the divergence of $\nu(y)\nabla\phi^*$, and $\nu(y)\,y = \frac{1}{2}Q'(y)$ by Eq. (9).

Theorem 3. *At $y = x\mu(x)$,*

$$\begin{aligned}&\boxed{g(x) + Q(y) = 2xy,} \\ &g'(x) = 2y, \qquad Q'(y) = 2x,\end{aligned} \tag{12}$$

so $Q(y) = \max_{x \ge 0}[2xy - g(x)]$: Q is the Legendre transform of g. Written in b, $g = x^2 - 2x - 2\ln b$ and $\mathcal{M} = 2x - 1 + 2b - b^2 + 2\ln b$: the logarithms are equal and opposite and cancel in $g + \mathcal{M}$. In spherical symmetry $|\nabla\hat{\phi}| = a_0\,\mu(x) < a_0$.

Proof. $g'(x) = 2x^2/(1 + x) = 2y$ and, by Eq. (9), $Q'(y) = 2y + 2(x - y) = 2x$. Then $\frac{d}{dx}[g + Q - 2xy] = 2y + 2x\,y' - 2y - 2x\,y' = 0$ and all three terms vanish at $x = 0$. Since $g'' > 0$ for $x > 0$, the stationary point $g'(x) = 2y$ of $2xy - g(x)$ is its unique maximum. Finally $|\nabla\hat{\phi}| = |\nabla\phi| - |\nabla\phi^*| = a_0(x - y) = a_0\mu(x)$. □

Corollary 2 (convexity and uniqueness). *The energy $W(q) = Q(|q|)$, $q = \nabla\phi^*/a_0$, is strictly convex for $q \neq 0$, with Hessian eigenvalues $Q'/y = 2\nu$ (twice) and $Q'' = 2\,dx/dy = 2(x + 1)^2/[x(x + 2)]$ (once), whose ratio*

$$\frac{1}{2} \le \frac{Q''}{Q'/y} = \frac{x + 1}{x + 2} < 1 \tag{13}$$

is the reciprocal of the AQUAL ratio $(x\mu)'/\mu = (x + 2)/(x + 1)$. On a bounded domain with given boundary values, Eq. (4) has at most one solution (ϕ^*, ϕ).

Proof. For a radial function the Hessian has eigenvalue Q'/y across $\hat{q}$ and Q'' along it. $Q'/y = 2x/y = 2\nu$ and $Q'' = 2/(dy/dx)$, both positive. Their ratio is $(dx/dy)(y/x) = 1/[(x\mu)'/\mu]$, since $y = x\mu$; this is the inverse Hessian of a Legendre pair. For uniqueness, the difference of two solutions of each Poisson equation in Eq. (4) is harmonic with zero boundary values, hence zero: first for ϕ^*, then for ϕ, whose source is then fixed. The

flux $\nu(y)\, y = x(y)$ is continuous at $\nabla\phi^* = 0$, where $x \simeq \sqrt{y}$, so the source $\nabla \cdot (\mathcal{M}'\nabla\phi^*)$ is well defined weakly. □

Where AQUAL degenerates at $\nabla\Phi = 0$ (both eigenvalues $\to 0$), the two-potential energy is singular there ($2\nu \to \infty$); the Legendre transform exchanges the two behaviours.

Summary
In nonrelativistic BIMOND the difference potential is Newtonian, independently of the interaction, exactly when $\alpha + \beta = 0$, and the tetrad form of the Einstein–Hilbert term gives that sign for a frame on the second root. Requiring the constitutive law $\mu(x) = x/(x+1)$ then fixes the interaction function uniquely as $\mathcal{M} = 2x - 1 + 2b - b^2 + 2\ln b$, which meets BIMOND's two limiting conditions with unit coefficient. It is the Legendre dual of the AQUAL kinetic function, the logarithms of the two cancelling in the pairing, and its energy is strictly convex with the reciprocal Hessian ratio.

References

[1] M. Milgrom, *Bimetric MOND gravity*, Phys. Rev. D **80** (2009) 123536, arXiv:0912.0790.

[2] M. Milgrom, *Matter and twin matter in bimetric MOND*, arXiv:1001.4444 (2010).

[3] M. Milgrom, *Quasi-linear formulation of MOND*, Mon. Not. R. Astron. Soc. **403** (2010) 886, arXiv:0911.5464.

[4] J. Bekenstein and M. Milgrom, *Does the missing mass problem signal the breakdown of Newtonian gravity?*, Astrophys. J. **286** (1984) 7.

[5] S. Nelson, *The energy functional of the constitutive law* $\mu(x) = x/(x+1)$, Field Dynamics Research (2026).

[6] S. Nelson, *The tetrad scheme and the two bounded regions*, Field Dynamics Research (2026).